

Paul's Online Notes

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You appear to be on a device with a "narrow" screen width (*i.e.* you are probably on a mobile phone). Due to the nature of the mathematics on this site it is best viewed in landscape mode. If your device is not in landscape mode many of the equations will run off the side of your device (you should be able to scroll/swipe to see them) and some of the menu items will be cut off due to the narrow screen width.

Section 12.7 : Calculus With Vector Functions

In this section we need to talk briefly about limits, derivatives and integrals of vector functions. As you will see, these behave in a fairly predictable manner. We will be doing all of the work in $\mathbb{R}^3$ but we can naturally extend the formulas/work in this section to $\mathbb{R}^n$ (*i.e.* n -dimensional space).

Let's start with limits. Here is the limit of a vector function.

$$\begin{aligned}\lim_{t \rightarrow a} \vec{r}(t) &= \lim_{t \rightarrow a} \langle f(t), g(t), h(t) \rangle \\ &= \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle \\ &= \lim_{t \rightarrow a} f(t) \vec{i} + \lim_{t \rightarrow a} g(t) \vec{j} + \lim_{t \rightarrow a} h(t) \vec{k}\end{aligned}$$

So, all that we do is take the limit of each of the component's functions and leave it as a vector.

Example 1 Compute $\lim_{t \rightarrow 1} \vec{r}(t)$ where $\vec{r}(t) = \left\langle t^3, \frac{\sin(3t - 3)}{t - 1}, e^{2t} \right\rangle$.

Hide Solution ▼

There really isn't all that much to do here.

$$\begin{aligned}
 \lim_{t \rightarrow 1} \vec{r}(t) &= \left\langle \lim_{t \rightarrow 1} t^3, \lim_{t \rightarrow 1} \frac{\sin(3t - 3)}{t - 1}, \lim_{t \rightarrow 1} e^{2t} \right\rangle \\
 &= \left\langle \lim_{t \rightarrow 1} t^3, \lim_{t \rightarrow 1} \frac{3 \cos(3t - 3)}{1}, \lim_{t \rightarrow 1} e^{2t} \right\rangle \\
 &= \langle 1, 3, e^2 \rangle
 \end{aligned}$$

Notice that we had to use **L'Hospital's Rule** on the y component.

Now let's take care of derivatives and after seeing how limits work it shouldn't be too surprising that we have the following for derivatives.

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t) \vec{i} + g'(t) \vec{j} + h'(t) \vec{k}$$

Example 2 Compute $\vec{r}'(t)$ for $\vec{r}(t) = t^6 \vec{i} + \sin(2t) \vec{j} - \ln(t + 1) \vec{k}$.

Hide Solution ▼

There really isn't too much to this problem other than taking the derivatives.

$$\vec{r}'(t) = 6t^5 \vec{i} + 2 \cos(2t) \vec{j} - \frac{1}{t + 1} \vec{k}$$

Most of the basic facts that we know about derivatives still hold however, just to make it clear here are some facts about derivatives of vector functions.

Facts

$$\begin{aligned}
\frac{d}{dt}(\vec{u} + \vec{v}) &= \vec{u}' + \vec{v}' \\
(c\vec{u})' &= c\vec{u}' \\
\frac{d}{dt}(f(t)\vec{u}(t)) &= f'(t)\vec{u}(t) + f(t)\vec{u}'(t) \\
\frac{d}{dt}(\vec{u} \cdot \vec{v}) &= \vec{u}' \cdot \vec{v} + \vec{u} \cdot \vec{v}' \\
\frac{d}{dt}(\vec{u} \times \vec{v}) &= \vec{u}' \times \vec{v} + \vec{u} \times \vec{v}' \\
\frac{d}{dt}(\vec{u}(f(t))) &= f'(t)\vec{u}'(f(t))
\end{aligned}$$

There is also one quick definition that we should get out of the way so that we can use it when we need to.

A **smooth curve** is any curve for which $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq \vec{0}$ for any t except possibly at the endpoints. A helix is a smooth curve, for example.

Finally, we need to discuss integrals of vector functions. Using both limits and derivatives as a guide it shouldn't be too surprising that we also have the following for integration for indefinite integrals

$$\begin{aligned}
\int \vec{r}(t) dt &= \left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle + \vec{c} \\
\int \vec{r}(t) dt &= \int f(t) dt \vec{i} + \int g(t) dt \vec{j} + \int h(t) dt \vec{k} + \vec{c}
\end{aligned}$$

and the following for definite integrals.

$$\begin{aligned}
\int_a^b \vec{r}(t) dt &= \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle \\
\int_a^b \vec{r}(t) dt &= \int_a^b f(t) dt \vec{i} + \int_a^b g(t) dt \vec{j} + \int_a^b h(t) dt \vec{k}
\end{aligned}$$

With the indefinite integrals we put in a constant of integration to make sure that it was clear that the constant in this case needs to be a vector instead of a regular constant.

Also, for the definite integrals we will sometimes write it as follows,

$$\int_a^b \vec{r}(t) dt = \left(\left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle \right) \Big|_a^b$$

$$\int_a^b \vec{r}(t) dt = \left(\int f(t) dt \vec{i} + \int g(t) dt \vec{j} + \int h(t) dt \vec{k} \right) \Big|_a^b$$

In other words, we will do the indefinite integral and then do the evaluation of the vector as a whole instead of on a component by component basis.

Example 3 Compute $\int \vec{r}(t) dt$ for $\vec{r}(t) = \langle \sin(t), 6, 4t \rangle$.

Hide Solution ▼

All we need to do is integrate each of the components and be done with it.

$$\int \vec{r}(t) dt = \langle -\cos(t), 6t, 2t^2 \rangle + \vec{c}$$

Example 4 Compute $\int_0^1 \vec{r}(t) dt$ for $\vec{r}(t) = \langle \sin(t), 6, 4t \rangle$.

Hide Solution ▼

In this case all that we need to do is reuse the result from the previous example and then do the evaluation.

$$\begin{aligned} \int_0^1 \vec{r}(t) dt &= \left(\langle -\cos(t), 6t, 2t^2 \rangle \right)_0^1 \\ &= \langle -\cos(1), 6, 2 \rangle - \langle -1, 0, 0 \rangle \\ &= \langle 1 - \cos(1), 6, 2 \rangle \end{aligned}$$